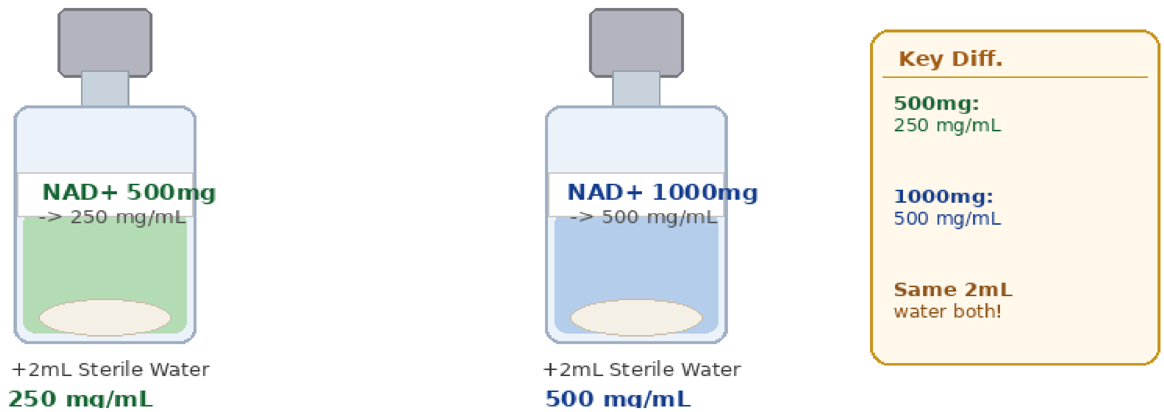
# NAD+ (Nicotinamide Adenine Dinucleotide)

**Complete Injection Guide — 500mg & 1000mg Vials**

Purity 99.8% | Endotoxin-Free | Heavy Metal-Free

## What You Have — Two Vial Options




Both vials use 2mL sterile water. 500mg gives 250mg/mL. 1000mg gives 500mg/mL. Same dose = different volume. Always check your vial before drawing.

## Product Specifications

| Parameter | 500mg Vial | 1000mg Vial |
|---|---|---|
| **Content** | 500mg lyophilized NAD+ | 1000mg lyophilized NAD+ |
| **Reconstitution** | 2mL sterile water | 2mL sterile water |
| **Concentration** | 250 mg/mL | 500 mg/mL |
| **Route** | IV drip (preferred) / SubQ | IV drip (preferred) / SubQ |
| **Purity** | 99.8% | 99.8% |
| **Endotoxins** | Non-detected (<0.1 EU/mg) | Non-detected |
| **Storage (unrecon.)** | 2-8°C refrigerated | 2-8°C refrigerated |
| **Storage (reconstituted)** | 2-8°C, use within 24 hours | 2-8°C, use within 24 hours |
| **Appearance** | Clear, slightly yellow solution | Same |

## SECTION 1 — What is NAD+ and Why Use It?

### About NAD+

NAD+ (Nicotinamide Adenine Dinucleotide) is a coenzyme found in every living cell. It is essential for energy metabolism (ATP production), DNA repair, mitochondrial function, and cellular longevity. NAD+ levels decline by up to 50% between ages 40-60, contributing to fatigue, cognitive decline, and accelerated aging. IV or SubQ supplementation rapidly restores cellular NAD+ levels, with significantly higher bioavailability than oral supplementation.

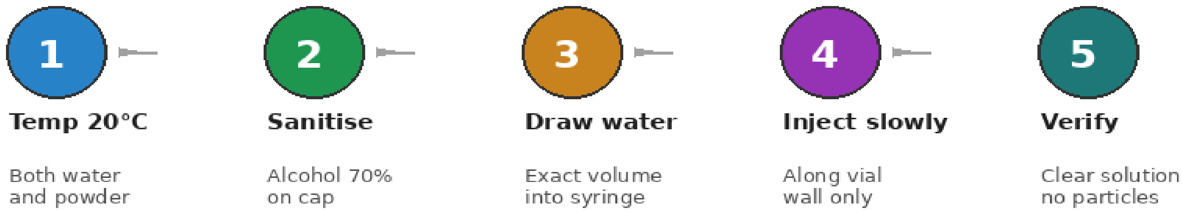
### Key Research Benefits

| Body System | Research-Supported Effects |
| --- | --- |
| **Energy & Metabolism** | Drives ATP production in mitochondria. Restores cellular energy, reduces fatigue and improves physical performance. |
| **DNA Repair** | Activates PARP enzymes and sirtuins (SIRT1-7) that repair DNA damage, protecting against cellular aging and mutation. |
| **Brain & Cognition** | Supports neuronal health, improves mental clarity, focus and memory. Research shows potential in Alzheimer's and Parkinson's models. |
| **Anti-Aging (Sirtuins)** | Activates longevity proteins (sirtuins) that regulate metabolism, stress resistance, and cellular lifespan. |
| **Cardiovascular** | Improves mitochondrial function in cardiac cells, reduces oxidative stress, supports heart health and circulation. |
| **Addiction & Mental Health** | Used in clinical IV protocols for addiction recovery, anxiety and depression. Replenishes depleted neurotransmitter precursors. |
| **Inflammation** | Regulates NF-kB signalling, reducing chronic inflammation at the cellular level. Supports immune system balance. |

**Why IV over oral?** Oral NAD+ supplements (NR, NMN) must be converted and have ~5-10% bioavailability. IV administration delivers NAD+ directly into circulation at 100% bioavailability, producing immediate cellular effects.

## SECTION 2 — Reconstitution Protocol

### 5-Step Reconstitution Process (Both Vials)




Follow all 5 steps in order. Use exactly 2mL sterile water for BOTH vials.

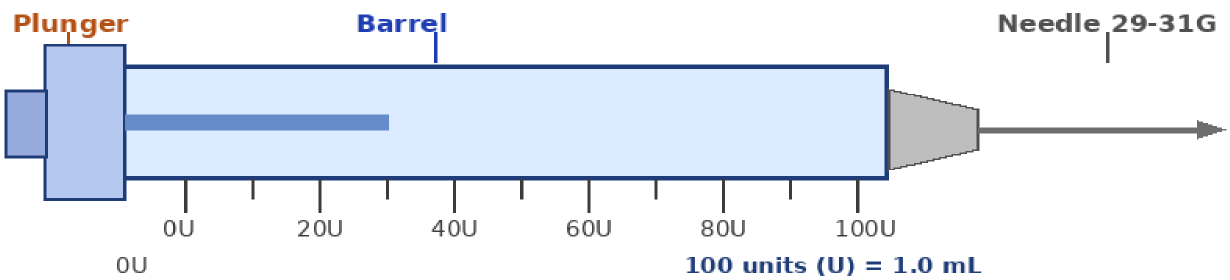
### Reconstitution Details

| Step | Action | Full Detail |
|---|---|---|
| 1 | **Set temp to 20°C** | Remove sterile water and NAD+ vial from fridge. Leave at room temperature 15-20 minutes. Cold mixing causes incomplete dissolution. |
| 2 | **Sanitise cap** | Wipe rubber stopper of BOTH vials with separate alcohol swabs. Let air-dry 15 seconds. Do not touch after wiping. |
| 3 | **Draw 2mL water** | Use a 3mL or 5mL syringe. Insert needle into sterile water vial, invert, pull back plunger to exactly 2mL. Remove air bubbles. |
| 4 | **Inject along wall** | Insert needle into NAD+ vial. Tilt so needle tip touches inside glass wall. Push SLOWLY — let water run down the wall onto the powder. Do NOT spray directly onto powder (causes foaming and degradation). |
| 5 | **Swirl — no shaking** | Gently roll vial between palms or swirl in circles. NAD+ dissolves in 30-60 seconds. Solution should be clear with faint yellow tint. DISCARD if cloudy, pink, or contains visible particles. |

**After reconstitution:** Label vial with date and time. Store at 2-8°C. Use within **24 hours** — NAD+ degrades faster than most peptides once dissolved. Do NOT freeze reconstituted solution.

## SECTION 3 — Dose Calculation

### How to Read the Syringe (100U = 1.0 mL)




Each 10 units (U) = 0.10 mL. 100U = 1.0 mL full volume.

### Dose Calculation Formula

| DOSE (mg) ------------------ CONC. (mg/mL) | = | VOLUME (mL) | x 100 | = UNITS (U) on syringe |
|---|---|---|---|---|

**Example — 500mg vial (250mg/mL):** 250mg dose → 250/250 = 1.0mL = 100U | 100mg dose → 100/250 = 0.40mL = 40U

**Example — 1000mg vial (500mg/mL):** 250mg dose → 250/500 = 0.50mL = 50U | 500mg dose → 500/500 = 1.0mL = 100U

| 500mg Vial (250 mg/mL after 2mL water) | | | |
|---|---|---|---|
| **Dose (mg)** | **Volume (mL)** | **Units (U)** | **Research Context** |
| **100 mg** | 0.40 mL | 40 U | Low / introductory dose |
| **250 mg** | 1.00 mL | 100 U | Standard dose — most protocols |
| **500 mg** | 2.00 mL | 200 U | Full vial — high dose protocol |

| 1000mg Vial (500 mg/mL after 2mL water) | | | |
|---|---|---|---|
| **Dose (mg)** | **Volume (mL)** | **Units (U)** | **Research Context** |
| **250 mg** | 0.50 mL | 50 U | Low / introductory dose |
| **500 mg** | 1.00 mL | 100 U | Standard dose — most protocols |
| **750 mg** | 1.50 mL | 150 U | High dose — advanced protocol |
| **1000 mg** | 2.00 mL | 200 U | Full vial — intensive protocol |

**Most common starting dose: 250mg.** First-time users should always start at 250mg or less to assess tolerance before increasing.

## SECTION 4 — Administration: IV Drip & SubQ Injection

### Infusion Rate — The Most Critical Rule

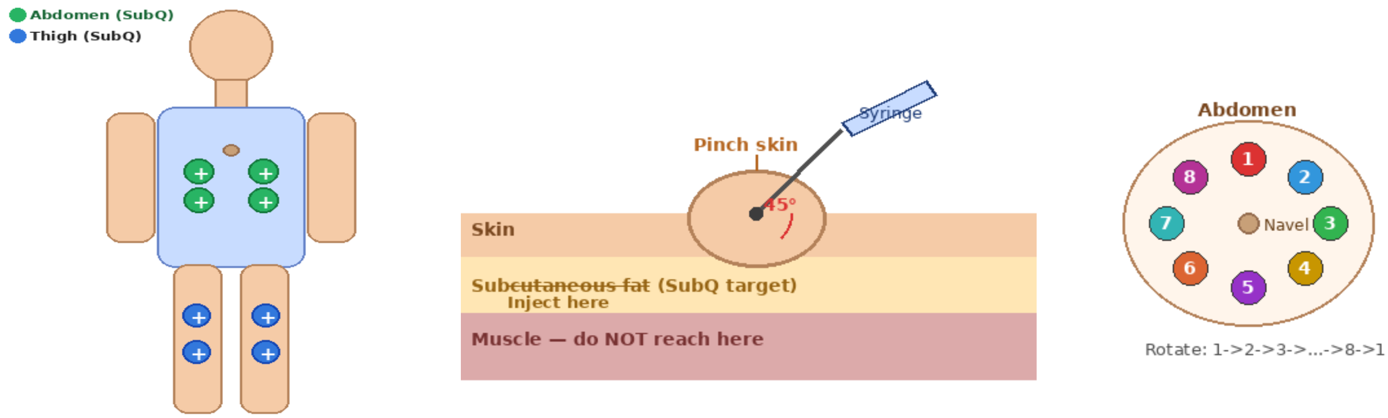

### IV Drip Protocol (Recommended Method)

| Step | Action | Detail |
|---|---|---|
| 1 | **Prepare IV bag** | Add reconstituted NAD+ to 100-250mL Normal Saline (NS 0.9%) IV bag. Mix gently. Cover bag with foil if light-sensitive. |
| 2 | **Set drip rate** | Set for 2-4 hour infusion. Use IV drip calculator: 250mL over 4hrs = ~62 drops/min (standard drip set). |
| 3 | **Monitor first 15 min** | Stay with patient. Watch for flushing, chest tightness, or nausea. These mean: SLOW DOWN the drip rate — do not stop entirely. |
| 4 | **Adjust if side effects** | If nausea or flushing occur: reduce drip rate by half. Side effects always resolve when rate is reduced. |
| 5 | **Flush line after** | After NAD+ completes, flush IV line with 20-30mL plain NS before removing or switching to another infusion. |

**Supplies needed for IV:** IV bag (NS 0.9%, 100-250mL), IV line/tubing, IV cannula (20-22G), alcohol swabs, gloves, drip stand, tape.

### SubQ Injection Method (Alternative)

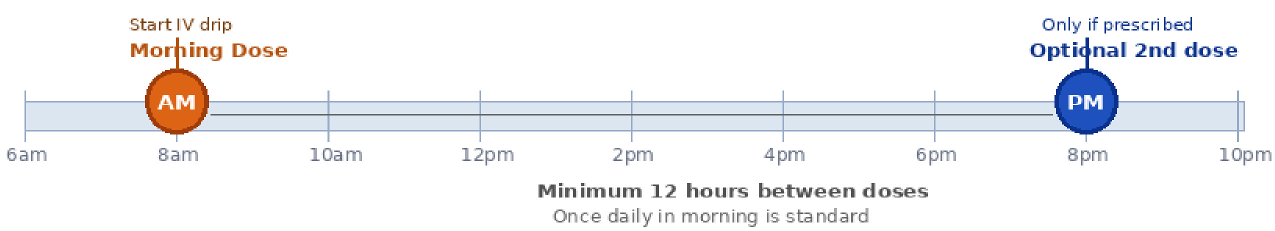
If IV administration is not available, NAD+ can be given subcutaneously. SubQ absorption is slower but effective for maintenance dosing. Maximum 1mL per injection site — split larger doses across multiple sites.




Left: SubQ injection sites (green=abdomen, blue=thigh). Centre: Pinch skin, insert at 45°, push slowly. Right: Rotate 8 sites in order — never repeat same spot.

## SECTION 5 — Dosing Protocol & Weekly Schedule

### Recommended Daily Timing




Morning infusion is standard. A second evening dose is optional and only if prescribed.

### Research Protocols by Goal

| Goal | Dose | Frequency | Duration | Notes |
|---|---|---|---|---|
| **General wellness & energy** | 250-500 mg | 2x per week | 4-8 weeks | Good starting protocol |
| **Anti-aging & longevity** | 500 mg | 2-3x per week | 8-12 weeks | Maintain with 1x/week after |
| **Cognitive enhancement** | 250-500 mg | 2x per week | 6-8 weeks | Can repeat cycles |
| **Addiction recovery** | 500-1000 mg | Daily x 10 days | 10 days | Clinical supervision required |
| **Athletic recovery** | 250-500 mg | 1-2x per week | Ongoing | Best taken day after training |
| **Rest / Off period** | — | — | 4 weeks | Allow 4 weeks before new cycle |

## SECTION 6 — Side Effects, Safety & Storage

### Supply Duration Calculator

| Vial | Dose per session | Duration (2x/week) |
|---|---|---|
| **500mg vial** | 250 mg (standard) | ~1 week per vial |
| **500mg vial** | 500 mg (full vial) | ~3-4 days per vial |
| **1000mg vial** | 250 mg (standard) | ~2 weeks per vial |
| **1000mg vial** | 500 mg (standard) | ~1 week per vial |

**Note:** Reconstituted NAD+ must be used within 24 hours. Store unreconstituted powder at 2-8°C refrigerated.

### Side Effects — Almost Always Rate-Related

NAD+ IV side effects are almost exclusively caused by infusing too fast. Slowing the drip rate resolves 95% of reactions within minutes.

| Side Effect | Cause | Severity | Solution |
|---|---|---|---|
| **Flushing / warmth** | Too fast infusion | Mild | Slow drip rate immediately |
| **Chest tightness** | Too fast infusion | Mild-Mod | Slow drip — monitor closely |
| **Nausea** | Too fast infusion | Mild | Slow drip; small meal before |
| **Headache** | Rate or dehydration | Mild | Hydrate well before infusion |
| **Palpitations** | Too fast infusion | Mild | Slow drip immediately |
| **Injection site redness** | SubQ — normal | Mild | Rotate sites each time |
| **Allergic reaction** | Rare | Serious | STOP immediately — seek help |

**Golden rule for NAD+ IV:** If any side effect occurs — slow the drip, do not stop it. Stopping abruptly and restarting can worsen symptoms. Simply reduce the rate and all effects resolve within 5-10 minutes.

### When to STOP and Seek Medical Attention

- Severe chest pain or pressure that does not resolve after slowing drip — STOP and call emergency
- Allergic reaction: hives, difficulty breathing, or facial swelling — EMERGENCY
- Fever above 38°C or signs of infection at IV/injection site
- Solution is cloudy, discoloured, or contains particles — DISCARD immediately
- Vomiting that prevents staying hydrated — discontinue and rehydrate

### Storage Summary

| State | Temperature | Max Duration | Notes |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **Lyophilized powder (unreconstituted)** | 2-8°C refrigerated | Per expiry date | Keep dry; do not freeze |
| **Reconstituted solution** | 2-8°C refrigerated | 24 hours ONLY | Label with time; degrade fast |
| **In IV bag (with NS)** | 2-8°C or use immediately | Within 4 hours | Infuse same day — no storage |
| **BAC/Sterile Water opened** | 2-8°C refrigerated | 28 days | Label with open date |

## 10 Golden Rules for NAD+ Administration

| # | Rule |
|---|---|
| **1** | SLOW is SAFE — never rush a NAD+ infusion. 2-4 hours minimum. |
| **2** | Always start at 250mg or less for first-time infusions. |
| **3** | If any side effect occurs — slow the drip, do not stop it. |
| **4** | Use reconstituted NAD+ within 24 hours — it degrades quickly. |
| **5** | Store unreconstituted powder at 2-8°C refrigerated. |
| **6** | Hydrate well before and during infusion — drink 500mL water beforehand. |
| **7** | Never infuse cold solution — always bring to room temperature first. |
| **8** | For SubQ: use a new needle every injection, rotate sites 1 through 8. |
| **9** | Label every reconstituted vial with date and time prepared. |
| **10** | Discard any solution that is cloudy, pink, or contains particles. |

**DISCLAIMER:** This guide is for informational and research reference purposes only. NAD+ is Research Use Only (RUO). IV administration should only be performed by or under the supervision of a qualified medical professional. is not responsible for misuse of this product.